

# Operational Definitions of the ARC Theory

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*The source of truth for the ARC Theory's notation: every equation the programme relies upon for Laws I to III, ranked by how much depends on it, each symbol defined and paired with its measurement; the notation of Laws IV and V is fixed in their mathematics until it joins.*

Working Paper v1.9.0 | First published 22 August 2026, revised 4 October 2026

## ABSTRACT

This document fixes what each symbol in the ARC Theory's equations means and how it would be measured. It ranks the twenty-one equations the programme relies upon for Laws I to III by how much depends on each, from the ARC Equation of Law I,  $U = I \times R^\alpha$ , through Law II's criterion,  $\beta_C > k$ , and Law III's ceiling,  $\alpha_{\text{crit}} = 1 / (1 - \gamma)$ , to the statistical methods the experiments use. It defines thirty-eight symbols in plain words, pairs each with the measurement that would produce it, names the nine letters the corpus uses for more than one quantity, and records the seven forms the programme has retracted, with the reason each was wrong. It reports no result and asserts no value: every figure it quotes is carried from the paper that measured it. The theory has five laws. Laws I to III were registered on 8 September 2026; Laws IV and V, the ARC Persistence Law and the ARC Embedding Law, were added on 24 September 2026, their propositions registered and neither law yet registered as a law or tested. Their notation is fixed in section 1 of their mathematics, version 1.0 (3 October 2026), and joins this document at its next version.

## How to read this

An exponent is a growth-shape number, not a size. It answers: when I double the effort, what happens to the result? An exponent of 1 means the result doubles too. An exponent of 2 means it quadruples. An exponent below 1 means you get less than double, which is the shape of diminishing returns. Every claim in this programme is a claim about the value of one of these shape numbers, so every argument turns on measuring one correctly.

**The one rule.** A symbol means what this file says it means, and nothing else. Where the same letter is used for two different quantities anywhere in the corpus, that letter appears in the collisions list below, and both readings are named. If you meet a symbol in a paper

and cannot find it here, that is a defect in the paper, not in your reading.

What the word law is doing here. The theory's five laws are named conjectures. Laws I to III were registered on 8 September 2026; Laws IV and V were added on 24 September 2026, their propositions registered and neither law yet registered as a law or tested. Each law states a relation held open to measurement; none is established, and the status line under every equation says which of definition, derivation, criterion, modelling choice or open measurement it is. Naming a conjecture is how it becomes arguable, not how it becomes true.

## The equations, ranked

Ranked by how much fails if the equation is wrong, not by how often it appears. The first four are the programme; the rest support them. The third column is the ranking rationale, so the order can be argued with rather than taken on trust. Everything else about each equation, what it says in plain words, what it is for, every symbol with its measurement, and what would refute it, is in its own section below and is not repeated here.

| # | EQUATION                                                                        | WHY IT RANKS HERE: WHAT FAILS IF IT IS WRONG                                                                                                                   |
|---|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $U = I \times R^\alpha$<br>The ARC Equation                                     | Nothing else in the programme survives. This is the frame, not a result.                                                                                       |
| 2 | $\beta_C > k$<br>The ARC Co-Scaling criterion                                   | The safety claim dies, and EQ-02 loses its operational meaning again, because the word stable in the ARC Bound is defined by this criterion.                   |
| 3 | $\alpha_{\text{crit}} = 1 / (1 - \gamma)$<br>The ARC Ceiling                    | The value two loses its support, though the law itself is separable: a different measured gamma yields a different ceiling and the structure stands.           |
| 4 | $\alpha \leq 2$<br>The ARC Bound                                                | The headline claim dies. The framework survives, because EQ-01 and EQ-04 do not depend on the value two.                                                       |
| 5 | $B(R) / A_c(R) \sim R^{\alpha(1 - \gamma) - 1}$<br>The burden-to-capacity ratio | The ceiling loses its derivation and the value two loses its support.                                                                                          |
| 6 | $\alpha = d / (d + 1), d = 2 / (1 + 3w)$<br>The dimensional derivation          | The cross-domain derivation dies; the empirical programme is untouched.                                                                                        |
| 7 | $f(xy) = f(x) * f(y)$<br>Cauchy's multiplicative functional equation            | The claim that a power law is forced rather than fitted collapses.                                                                                             |
| 8 | $\alpha = \ln(E1 / E2) / \ln(R2 / R1)$<br>The slope estimator                   | Every measured number in the programme is wrong, including the retraction. Operationally this is the most important equation here, whatever its rank as a law. |

| #  | EQUATION                                                                                                           | WHY IT RANKS HERE: WHAT FAILS IF IT IS WRONG                                                                  |
|----|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 9  | resample with replacement,<br>refit, take the 2.5th and 97.5th<br>percentiles<br>The bootstrap confidence interval | Every stated interval is wrong, including the one that withdrew<br>2.24.                                      |
| 10 | $E \sim R^{(-1/2)}$ , that is $\alpha = 0.5$<br>The zero-parameter null                                            | Nothing of the framework falls. If it is RIGHT, the compounding<br>reading of the measurement is unnecessary. |
| 11 | $P(H   E) = P(E   H) * P(H) / P(E)$<br>Bayes' theorem                                                              | Nothing. It is a theorem.                                                                                     |
| 12 | $d = (\text{mean1} - \text{mean2}) / \text{standard deviation}$<br>Cohen's d                                       | Nothing. It is a definition.                                                                                  |
| 13 | $X^2 = -2 * \sum(\ln p_i)$ , with 2k<br>degrees of freedom<br>Fisher's method for combining p-values               | The combined figure falls; the individual results are untouched.                                              |
| 14 | $\alpha = 1 / (1 - \beta_L)$<br>The self-reference theorem                                                         | Alpha becomes a fitted parameter with no derivation, and the<br>programme becomes curve-fitting.              |
| 15 | $dg/dr = a * g^{(\beta_L)}$<br>The growth equation                                                                 | EQ-05 loses its derivation.                                                                                   |
| 16 | $E(R) = E_0 * R^{(-\alpha)}$<br>Error decay                                                                        | The link between the frame and the data breaks.                                                               |
| 17 | $A(C) = A_{\max} * C^{\beta_H} / (C^{\beta_H} + C_s^{\beta_H})$<br>The saturation curve                            | The saturating treatment in Paper X loses its functional form.                                                |
| 18 | $A(R) = A_0 * R^{\beta_A}$<br>Alignment scaling                                                                    | Paper II's alignment-side scaling claim dies.                                                                 |
| 19 | $k = \delta - 1/\alpha$<br>The self-acceleration exponent                                                          | The drift term in EQ-04 loses its derivation and must be measured<br>directly.                                |
| 20 | $\beta_X = (1 - \beta_L) / (1 - \beta_{L_{\text{prime}}})$<br>The boundary gap                                     | Paper XIII's cross-system comparison loses its measure.                                                       |
| 21 | $P/M = M^{(3/4)} / M = M^{(-1/4)}$<br>The metabolic analogy                                                        | An analogy is lost. No claim depends on it.                                                                   |

## Each one, explained

Each symbol is defined in full at its first appearance, in rank order, and later equations point back to that definition rather than repeating it. A definition restated four times reads as padding and gives a reader four chances to meet a version that has drifted.

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## 1 The ARC Equation LAW I, THE ARC PRINCIPLE

**In words.** A system's usable capability equals what it starts with, multiplied by how deeply it can improve itself, raised to a power.

$$U = I \times R^{\alpha}$$

**What it is for.** It is the definition that makes the rest measurable. It does not assert a value; it says which number to go and measure. Every other equation here is a statement about alpha.

**Worked.** Capability here is a latent, ratio-scaled measure with no ceiling, like the level a system reaches on a calibrated difficulty ladder, never a bounded percentage score: a score capped at 100 cannot follow this arithmetic, which is why the programme's own instruments measure on the ladder scale, after a bounded percentage score was found running through unbounded arithmetic and the worked example was recast on 27 August 2026. Suppose a system sits at capability  $U = 40$  on such a ladder at one pass, so  $I = 40$ . Allow it 16 passes, so  $R = 16$ . If  $\alpha = 0$ , looping is worthless:  $16^0 = 1$ , so  $U = 40$ , no gain. If  $\alpha = 0.5$ ,  $16^{0.5} = 4$ , so  $U = 160$ . If  $\alpha = 1$ ,  $16^1 = 16$ , so  $U = 640$ . If  $\alpha = 2$ ,  $16^2 = 256$ , so  $U = 10,240$ , and each loop makes the next more productive. That is the whole argument in four lines, and the question is which of those worlds we are in. The honest answer so far is the first or second.

**Measured as.** Varied: recursive depth  $R$ , counted on a configuration ladder fixed in advance. Read out: usable capability  $U$ , scored on a target battery frozen before the first measurement. Held fixed: the battery, the ladder and the scoring instrument, for the whole run.

**U** usable capability

What the system can actually do, after everything is accounted for. The output side of the equation.

**How it is measured.** Score on a fixed target battery held constant across the whole experiment. The battery must be frozen before any measurement, because a battery that changes during a run makes the exponent meaningless.

**Domain.** positive

**Units.** battery score, dimensionless once normalised

**I** base intelligence

What the system could do before any self-improvement. The starting point.

**How it is measured.** The same battery score measured at recursion depth one, before any improvement pass is applied.

**Domain.** positive

**Units.** same as  $U$

**R** recursive depth

How many times the system has been through its own improvement loop. If it reviews and revises its own work three times, R is three.

**How it is measured.** Count of completed improvement passes under a fixed configuration ladder. Must be counted, never estimated, and the ladder fixed in advance.

**Domain.** integer, at least one

**Units.** dimensionless count

$\alpha$  (alpha) the capability-growth exponent

The growth-shape number. It says what happens to capability when you double the recursive depth. At  $\alpha = 1$  capability doubles. At  $\alpha = 2$  it quadruples. Below 1 you get diminishing returns.

**How it is measured.** The slope of a straight line fitted to capability against recursive depth when both are plotted on logarithmic axes. In practice, EQ-07 applied across the measured range, with the interval reported always. EQ-07 is written on the error side, so using it for  $\alpha$  assumes the error-decay exponent is this same  $\alpha$ . Whether it is has never been stated anywhere in the corpus, and it is an open collision: see  $\alpha$  in the error-decay equation, below.

**Domain.** unbounded in principle; claimed at most two for systems that hold together

**Status.** definitional; the exponent is open to measurement, and the measured value is currently 0.49 with interval  $[-1.3, 2.9]$ , which distinguishes nothing

Relied upon in: foundational, executive-summary, paper-i-arc-principle, paper-c-pnp.

## 2 The ARC Co-Scaling criterion LAW II, THE ARC CO-SCALING LAW

**In words.** A system stays aligned with itself only while its power to correct grows faster than its tendency to drift.

$$\text{beta\_C} > k$$

**What it is for.** The safety criterion, and since 22 August 2026 also the meter that makes EQ-02 falsifiable: stability is no longer asserted, it is measured as correction out-scaling drift.

**Worked.** A system doubles in capability. Its correction capacity rises 30 per cent, so  $\text{beta\_C}$  is about 0.38. Its drift rises 20 per cent, so  $k$  is about 0.26. Since  $0.38 > 0.26$  correction is pulling ahead and the off-target fraction shrinks as it grows. Reverse the two and the same system grows more capable while getting harder to keep on target. Read the comparison, never either number alone:  $\text{beta\_C} = 0.9$  with  $k = 1.1$  is unstable despite far stronger correction.

**Read this before using it.** The  $\text{beta}$  in this criterion is NOT the  $\text{beta}$  in EQ-05. See collisions.

**Measured as.** Varied: recursive depth. Read out: the two exponents  $\beta_C$  and  $k$ , each as a log-log slope from the same estimator that yields  $\alpha$ . Held fixed: the drift and correction instruments, so that the two slopes are commensurable. The criterion is the comparison of the two, not a quantity fitted in its own right.

$\beta_C$  ( $\beta_C$ ) the correction exponent

How fast the brakes strengthen. The rate at which a system's power to correct itself grows as it gets more capable.

**How it is measured.** Log-log slope of corrective strength against capability, from the same estimator as  $\alpha$ , EQ-07.

**Domain.** real

$k$  the drift-acceleration exponent

How fast the engine speeds up. The rate at which a system's tendency to drift away from its target accelerates as it improves itself.

**How it is measured.** Log-log slope of drift against capability, same estimator; or derived from cost scaling through EQ-11.

**Domain.** real; negative values mean drift decelerates

**Status.** criterion; both exponents estimable from the same log-log slope estimator that yields  $\alpha$ , which is why one instrument serves both laws

Relied upon in: hrih-paper, paper-x-coupled-coscaling.

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### 3 The ARC Ceiling LAW III, THE ARC CEILING

**In words.** The ceiling on the growth-shape number is one divided by the correction shortfall, where the shortfall is how far the correction-leverage number falls short of one.

$$\alpha_{crit} = 1 / (1 - \gamma)$$

**What it is for.** It is where the number two comes from. Feed it one assumption, that  $\gamma$  is one half, and it returns two. Change the assumption and the ceiling moves with it.

**Conditional on class.** The ceiling this returns is conditional on the corrector's class, because its only input is the leverage exponent and that exponent is class-dependent. Feed it the same-class cap of one half and it returns two. Feed it a cross-class exponent above one half and it returns more than two. The equation is unchanged either way; what changes is which system you are asking about.

**Worked.** The equation is a reciprocal, so the smaller the shortfall the higher the ceiling.  $\gamma = 0.3$  gives a shortfall of 0.7 and a ceiling of 1.43.  $\gamma = 0.5$  gives 0.5 and exactly 2.00.  $\gamma = 0.8$  gives 0.2 and 5.00.  $\gamma = 0.9$  gives 0.1 and 10.00. Read the middle row: a corrector scaling as the square root of capability gives a ceiling of exactly two. That is where the famous number comes from, as an output under one assumption, not a discovery.

**Read this before using it.** The retracted form  $1/\gamma$  and this corrected form return the same answer, two, at exactly  $\gamma = \text{one half}$ . That single point of agreement is the programme's headline number, which is why the wrong form survived undetected on the glossary page for six days. Below one half the two forms move in opposite directions: the retracted form makes a WORSE corrector raise the ceiling, which inverts a bound into a floor.

**Measured as.** Not measured directly. Both sides are estimated separately, the correction-leverage exponent from correction accumulation and  $\alpha$  from EQ-07, and the relation is tested by comparing the two estimates rather than by fitting it.

$\alpha_{\text{crit}}$  ( $\alpha_{\text{crit}}$ ) the stability ceiling

The highest growth-shape number a system can have and still hold itself together.

**How it is measured.** Not measured directly. Computed from  $\gamma$  through EQ-03. To test it you measure  $\gamma$ , not the ceiling.

**Domain.** at least one

$\gamma$  ( $\gamma$ ) the correction-leverage exponent

How much good it does to correct yourself. It measures how effectively internal correction converts accumulated capability into actual improvement. Higher is better. Written  $\chi$  from the working paper of 16 August 2026 onward, because  $\gamma$  had carried three incompatible readings; the theory registration still says  $\gamma$  and the two are the same quantity. Use  $\chi$  in any new writing and  $\gamma$  only when quoting the registration.

**How it is measured.** The per-family slope of misalignment removed per review pass against checker capability, under a blinded scoring instrument, with the target battery held fixed. Drafted as study-ad; never measured on any real system.

**Domain.** assumed at most one half for SAME-CLASS correctors, under the independence premise. Not a universal cap: it is a cap on that class

**Class dependence.** The cap is a property of the corrector's CLASS, not of correction as such. A corrector sharing the system's substrate cannot be anti-correlated with itself, which is where the one half comes from. A corrector from a different class is not bound by it and may exceed it. Build order is a lever on class: as the field currently builds, a corrector added after training on the vast corpus is drawn from that corpus and is close to maximally same-class, whereas one built and validated first on a separate curated corpus has the opportunity to differ. Opportunity rather than guarantee: a curated corpus drawn from the same distribution would separate nothing.

**What this rests on.** The value one half is a conjecture and not a derivation. It rests on the premise that internally accumulated corrections combine like independent samples. This is the single undefended assumption on which the ceiling at two rests, and it is deliberately left open.

$\chi$  (chi) the correction-service elasticity in capability

How much more correction a system can deliver when it becomes more capable.

**How it is measured.** The elasticity of correction service with respect to capability. This is the same quantity the theory registration calls the correction-leverage exponent gamma; chi is the name used from the working paper of 16 August 2026 onward, because gamma had carried three incompatible readings. Not identifiable from a single path: needs crossed or off-path variation.

**Domain.** 0 to below 1 for a finite ceiling

**Status.** corrected 16 August 2026; see retracted\_forms

Relied upon in: foundational, executive-summary.

#### 4 **The ARC Bound** THE SAME-CLASS INSTANCE OF LAW III, THE ARC CEILING, AT GAMMA EQUAL TO ONE HALF

**In words.** The growth-shape number cannot exceed two and still describe a system that holds itself together.

$$\alpha \leq 2$$

**What it is for.** The programme's headline claim. Not an impossibility claim: the region above two is enterable, but not correctable.

**What it is a bound of.** The ARC Bound is the SAME-CLASS bound. It was never universal, because its derivation runs through a leverage cap that this estate has always stated as a same-class cap, but until 22 August 2026 no surface a reader would meet said so. A cross-class system measured above two therefore does not refute it; it tests the build-order prediction instead. The kill condition now requires the corrector's class to be declared and evidenced before the run, on the same footing as the measurement window, because otherwise the assertion that a corrector was cross-class becomes a fresh escape of exactly the shape that the metering of the same day closed.

**Not a separate law.** The ARC Bound is the ARC Ceiling's output under one assumption, not a law of its own. The famous two is this equation's answer, never its input. Ranked fourth for that reason, below the equation that produces it.

**Measured as.** Not measured directly. It is a threshold on alpha, so it is tested by measuring alpha through EQ-07 and asking whether the estimate and its interval sit at or below two.

$\alpha$  (alpha) the capability-growth exponent

Defined in full at EQ-01, its first use.

**Status.** conditional under one stated assumption still on trial, that same-class correctors accumulate corrections the way independent samples do; refutable since 22 August 2026, when all four terms of its kill condition were metered: the exponent by its interval excluding two rather than its point estimate, the window by prior declaration, the system by a genuine self-referential loop, and stability by the ARC Co-Scaling Law's condition

Relied upon in: foundational, executive-summary.

## 5 The burden-to-capacity ratio THE DERIVATION BEHIND LAW III

**In words.** Compare how fast a system's need for correction is growing against how fast its ability to correct is growing. Whether that gap opens or closes decides everything.

$$B(R) / A_c(R) \sim R^{\alpha(1 - \gamma) - 1}$$

**What it is for.** This is where the ceiling actually comes from. The ceiling equation is its consequence, not its premise.

**Worked.** Everything hinges on the sign of  $\alpha(1 - \gamma) - 1$ . Negative, when  $\alpha$  is below the ceiling: the ratio shrinks, the corrector catches up, stable. Positive: the ratio grows without limit, uncorrected error piles up, unstable. Exactly zero: the exponents tie and equality is NOT automatically safe, because the outcome then turns on coefficients, delays, starting conditions and saturation, none of which the exponent captures. Set it to zero and solve and you get the ceiling law: that is where Law III comes from.

**What would prove it wrong.** Its premise, that correction burden tracks the capability a system is ADDING. If burden instead tracks the capability it already HAS, the growth exponent cancels out of the stability condition entirely and no ceiling on growth survives at all. Law II would survive; Law III would not.

**Read this before using it.** The capacity term is written  $A_c$  here, not a bare  $A$ . The papers write it as  $A$ , which already means alignment elsewhere in this corpus. Reusing the bare letter while auditing letter reuse would have been the same mistake in a new place, so the subscript is introduced here and the papers should adopt it.

**Measured as.** Not measured directly. What matters is the sign of the exponent, and that is read off two separately estimated quantities,  $\alpha$  from EQ-07 and the correction-leverage exponent from correction accumulation. The exactly-zero case is not automatically safe and is reported rather than rounded away.

**B** correction burden

How much correcting a system needs. It grows as the system does more, because more capability means more that can go wrong.

**How it is measured.** Volume of correction required per unit of capability added, measured over the same fixed battery used for capability.

**Domain.** non-negative

**Units.** correction units, dimensionless once normalised

**A\_c** correction capacity

How much correcting a system can actually do. The other side of the burden.

**How it is measured.** Correction delivered per pass at a given capability, from the same blinded instrument that supplies the correction exponent.

**Domain.** positive

**Units.** same as burden, so the ratio is dimensionless

**R** recursive depth

Defined in full at EQ-01, its first use.

$\alpha$  (alpha) the capability-growth exponent

Defined in full at EQ-01, its first use.

$\gamma$  (gamma) the correction-leverage exponent

Defined in full at EQ-03, its first use.

**Status.** derivation; the premise is stated and is the single point of attack

Relied upon in: foundational.

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## 6 The dimensional derivation

**In words.** The growth-shape number equals the effective dimension divided by that dimension plus one, and the dimension itself follows from an equation-of-state parameter.

$$\alpha = d / (d + 1), \quad d = 2 / (1 + 3w)$$

**What it is for.** The route by which the framework connects to a physical derivation rather than a fitted one.

**Measured as.** Not measured on any system.  $d$  and  $w$  enter from the physical derivation as a consistency route, not as fitted quantities.

$\alpha$  (alpha) the capability-growth exponent

Defined in full at EQ-01, its first use.

$d$  effective dimension and equation-of-state parameter

$w$  effective dimension and equation-of-state parameter

In the dimensional derivation,  $d$  is the effective number of directions the system can grow in, and  $w$  is the parameter that fixes it.

**How it is measured.** Not measured on any AI system. These come from the physical derivation and enter the framework as a consistency route, not as fitted quantities.

**Domain.**  $d$  positive

**Status.** derivation within stated assumptions

Relied upon in: on-the-origin-of-scaling-laws.

## 7 Cauchy's multiplicative functional equation

**In words.** If doubling the input always multiplies the output by the same factor no matter where you started, the relationship can only be a power law and nothing else fits.

$$f(xy) = f(x) * f(y)$$

**What it is for.** The argument that scaling laws must be power laws rather than merely happen to look like them.

**What would prove it wrong.** A continuous function satisfying the equation that is not a power law. Mathematically foreclosed under the stated regularity conditions, which is why the empirical route matters more: real domains that violate the classification.

**Measured as.** Not measured. It is a classical result, and what has to be checked in each application is whether the regularity conditions hold over the domain claimed.

$f$  the unknown function and its arguments, in Cauchy's equation

$x$  the unknown function and its arguments, in Cauchy's equation

$y$  the unknown function and its arguments, in Cauchy's equation

A rule you do not yet know, and two numbers you feed it. The equation asks which rules turn multiplication of the inputs into multiplication of the outputs; power laws are the answer.

**How it is measured.** Standard mathematical notation for the multiplicative functional equation.

**Domain.** positive reals in the form used here

**Status.** classical result; the content is in whether the regularity conditions hold in the domains claimed

Relied upon in: paper-vii-cauchy-unification.

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## 8 The slope estimator THE MEASUREMENT RECIPE

**In words.** To find the growth-shape number from data, take two measurements, divide one error by the other, take the logarithm, and divide by the logarithm of the ratio of the two depths.

$$\alpha = \ln(E1 / E2) / \ln(R2 / R1)$$

**What it is for.** This is the only equation in the programme that touches data. Every measured exponent anywhere in the corpus, including the retracted 2.24 and the corrected 0.49, came out of this estimator or its regression form.

**Worked.** A power law curves on ordinary axes and straightens under logarithms:  $\log U = \log I + \alpha \times \log R$ , which is a straight line  $y = c + mx$  whose slope IS  $\alpha$ . A logarithm answers how many times you multiply by ten to reach a number, so equal steps on the axis are equal multiplications and a multiplicative relationship becomes a line. What goes wrong: a slope fitted across a factor of two in R is nearly meaningless, slopes need about a decade to be trustworthy; curvature fitted as a slope still returns a number that means nothing; and a high r-squared says the points sit near the line, never that the line is the right model.

**Read this before using it.** The corpus writes this two ways: with a leading minus and the error ratio inverted, and without. They are the same estimator. A reader meeting both may reasonably think they are different equations; they are not.

**Measured as.** Varied: recursive depth, at two checkpoints R1 and R2 chosen before the run. Read out: error E1 and E2 at those two depths, from which the slope is computed. Held fixed: the error instrument and the ladder, because a slope taken between two differently measured points is not a slope.

$\alpha$  (alpha) the capability-growth exponent

Defined in full at EQ-01, its first use.

E error

How wrong the system is. The thing that is supposed to fall as it improves.

**How it is measured.** Error rate on the frozen target battery. Must be the same battery at every depth.

**Domain.** non-negative

**Units.** error rate, dimensionless

R recursive depth

Defined in full at EQ-01, its first use.

E1 (E1) the two measurement points

E2 (E2) the two measurement points

**R<sub>1</sub>** (R1) the two measurement points

**R<sub>2</sub>** (R2) the two measurement points

The pair of readings the slope is worked out from: error at each of two depths, and the two depths themselves. Two points make a line.

**How it is measured.** Error and recursive depth at the earlier and later of the two checkpoints entering the slope estimator.

**Domain.** positive

**Units.** E as error; R as depth

**Status.** standard log-log slope; the same estimator supplies  $\beta_C$  and  $k$  in EQ-04, which is what lets one instrument meter both laws

Relied upon in: paper-ii-experimental-validation, paper-iii-alignment-scaling-problem.

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## 9 The bootstrap confidence interval

**In words.** Rather than trusting one estimate, rebuild the dataset thousands of times by resampling it, and see how much the answer moves.

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resample with replacement, refit, take the 2.5th and 97.5th
percentiles
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**What it is for.** Every claim of significance in the programme. It is what produced the [-1.3, 2.9] interval that the point estimate must never be printed without.

**What would prove it wrong.** Nothing about the method. It is MISUSED by resampling data that are not independent, for instance repeated runs of one model treated as separate observations, which makes the interval look narrower than the truth.

**Measured as.** Not measured; it is the procedure that puts an interval on something else that was. Varied: the resample. Read out: the 2.5th and 97.5th percentiles of the refitted estimate. Held fixed: the dataset, the fitting procedure and the number of resamples, all declared before the first resample is drawn.

This equation carries no symbol. It is a procedure stated in words, so an empty symbol list here is a fact about the method and not an omission in the register.

**Status.** standard method; the risk is application, not theory

Relied upon in: paper-ii-experimental-validation.

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## 10 The zero-parameter null

**In words.** If each pass of reasoning is just an independent noisy attempt and errors average out, then plain statistics predicts an exponent of one half with no theory at all.

$$E \sim R^{(-1/2)}, \text{ that is } \alpha = 0.5$$

**What it is for.** It is the null this programme raised against ITSELF, and it is the most dangerous equation in the register. The programme's own corrected headline measurement is 0.49. That sits almost exactly on what averaging predicts with no framework at all.

**Worked.** Where the one half comes from, with no theory at all: the central limit theorem. Average  $n$  independent noisy attempts and the error shrinks in proportion to the square root of  $n$ , so the predicted exponent is 0.5 with no compounding, no leverage and no framework. The programme's own corrected measurement is 0.49. A measurement equally consistent with the framework and with plain averaging does not support either; it shows the measurement cannot tell them apart. Raised by this programme against itself, 8 August 2026, published 15 August.

**What would prove it wrong.** This condition fires AGAINST the framework, not for it. It fires, and the compounding reading is withdrawn, if a test built in advance to separate the two returns the null's prediction rather than a stated deviation from 0.5. It is retired only when such a test returns a deviation the framework predicted.

**Measured as.** Not measured; it is the null that a measurement is scored against. It fixes  $\alpha$  at one half with no fitted parameter, so a measured  $\alpha$  is compared with one half rather than with zero.

E error

Defined in full at EQ-07, its first use.

R recursive depth

Defined in full at EQ-01, its first use.

$\alpha$  (alpha) the capability-growth exponent

Defined in full at EQ-01, its first use.

**Status.** open and adverse; raised by this programme against its own headline number

Relied upon in: paper-ii-experimental-validation.

## 11 Bayes' theorem

**In words.** Start with how likely you thought something was, look at the new evidence, and adjust in proportion to how much better that evidence is explained by your idea being true than by it being false.

$$P(H | E) = P(E | H) * P(H) / P(E)$$

**What it is for.** All reasoning in the programme about priors and about what evidence would change them.

**Worked.** Worked, because this is the one people nod at without following. Suppose before any data you think there is a 30 per cent chance the correction exponent is one half, so  $P(H) = 0.3$ . A measurement arrives that would appear 80 per cent of the time if H is true and 20 per cent of the time if it is false. How often would you see it at all:  $P(E) = (0.8 \times 0.3) + (0.2 \times 0.7) = 0.24 + 0.14 = 0.38$ . Then  $P(H|E) = 0.8 \times 0.3 / 0.38 = 0.63$ . Belief moves from 30 per cent to 63 per cent. Notice it did not jump to certainty: the evidence multiplied the odds by four, from 3 to 7 up to 12 to 7, which moved the probability from 30 per cent only to 63 per cent, and odds and probability must never be quoted as though they were the same movement. That restraint is the point of using the rule rather than reasoning by impression. Operationalised here three ways: dated priors recorded in writing BEFORE any confirmatory data, with what would change them; kill conditions, which are commitments about  $P(E|H)$  written in advance, which is why the window must be declared before the run; and the reason consistent-with is not supports, because when  $P(E|H)$  and  $P(E|\text{not } H)$  are about equal the ratio is about one and the posterior barely moves from the prior.

**What would prove it wrong.** Nothing. It is provable from the definition of conditional probability, not an empirical claim. What can be wrong is its APPLICATION: priors chosen after the fact, likelihoods guessed to suit, or a claim of updating without recording what the prior was.

**Measured as.** Not measured. It is the rule by which a measurement moves a stated prior, and the discipline is that the prior is recorded before the evidence is seen.

H hypothesis and evidence, in Bayes' theorem

H is the claim being tested. E is what you observed. Bayes' theorem says how much the observation should move your confidence in the claim.

**How it is measured.** Standard statistical notation, not a quantity of this programme. Included so the equation can be read without another reference.

**Domain.** probabilities lie between zero and one

Also printed above, and defined elsewhere in this register: P in the P collision, which names the Bayes reading and the metabolic one, E in the E collision, and the H entry, which defines the evidence reading.

**Status.** theorem; the discipline is in recording the prior before the evidence

---

## 12 Cohen's d

**In words.** How far apart two groups are, measured in units of how spread out the data already is, so a difference can be called large or small without knowing what was measured.

$$d = (\text{mean1} - \text{mean2}) / \text{standard deviation}$$

**What it is for.** How big any measured difference in the programme actually is.

**Worked.** Conventionally about 0.2 is small, 0.5 moderate, 0.8 large, and the sign gives direction. A d of -0.53 means the first group scored lower by rather more than half a standard deviation. Dividing by the spread is what makes it portable: a 5-point gap is enormous if scores usually vary by 2 and trivial if they usually vary by 50.

**What would prove it wrong.** Nothing about the formula. It is MISAPPLIED when the two groups have very different spreads, so dividing by one shared standard deviation flattens or buries the difference, and when it is reported without the sample size that says how reliable it is.

**Read this before using it.** This d is not the programme's dimensional d. It belongs to statistics and must never be renamed; only the programme's own uses can take subscripts.

**Measured as.** Varied: group membership. Read out: the difference between the two group means. Held fixed: the measured quantity and its standard deviation, which is the unit the difference is expressed in.

**mean1** (mean1) the two group averages, in Cohen's d

**mean2** (mean2) the two group averages, in Cohen's d

The average of each of the two groups being compared, before the difference between them is expressed in standard deviations.

**How it is measured.** Standard statistical notation, not a quantity of this programme.

**Domain.** unbounded

**Units.** as for the measured quantity

Also printed above, and defined elsewhere in this register: d in the d collision, which names the effect-size reading.

**Status.** standard statistic

---

### 13 Fisher's method for combining p-values

**In words.** A way of pooling several separate weak results into one overall figure, which only works if the results were genuinely independent of one another.

$$X^2 = -2 * \sum(\ln p_i), \text{ with } 2k \text{ degrees of freedom}$$

**What it is for.** One published combined figure, since withdrawn.

**What would prove it wrong.** Nothing about the method. The live question is whether the independence condition holds, and here it does not, which is why the combined figure was withdrawn rather than defended.

**Measured as.** Withdrawn in application, so nothing is currently measured with it. Read out: a combined statistic pooled from k separately reported p-values. Held fixed: the requirement that the pooled results be genuinely independent, which was never established here.

$X^2$  (X) the combined test statistic and its inputs

$p_i$  the combined test statistic and its inputs

A way of pooling several separate tests into one verdict, and the individual results being pooled.

**How it is measured.** Standard statistical notation for Fisher's method, not a quantity of this programme. Included so the equation can be read without another reference.

**Domain.** p between zero and one

Also printed above, and defined elsewhere in this register: k in the k collision, which names the pooled-p-value reading.

**Status.** withdrawn in application; independence was never established

---

### 14 The self-reference theorem FOUNDATIONAL PAPER, THEOREM 2

**In words.** If a system's rate of improvement depends on how good it already is, raised to some power, then its capability grows as a power law whose exponent is one over the shortfall of that power from one.

$$\alpha = 1 / (1 - \beta_L)$$

**What it is for.** It is where alpha comes from mechanically rather than by fitting. It converts a statement about self-reference into a measurable exponent.

**Read this before using it.** This equation and EQ-03 share the shape one-over-one-minus-something. They are different laws with different arguments. Recognising the shape is not recognising the law.

**Measured as.** Not measured. It is a proof within stated assumptions, and its inputs are measured elsewhere: beta\_L from EQ-06 and alpha from EQ-07.

$\alpha$  (alpha) the capability-growth exponent

Defined in full at EQ-01, its first use.

$\beta_L$  (beta\_L) the self-referential coupling

How strongly a system's rate of improvement depends on how good it already is. At zero, improvement is constant regardless of skill. Near one, the better it gets the faster it gets better, and growth runs away.

**How it is measured.** Fitted from the growth equation EQ-06 over the measured range, then checked against alpha through EQ-05 as an independent consistency test. That check is an unexploited falsifier: measure beta\_L, predict alpha, compare with the fitted alpha.

**Domain.** less than one for finite growth

**Status.** proved within stated assumptions; not independently reviewed

Relied upon in: foundational, on-the-origin-of-scaling-laws, paper-ii-experimental-validation, paper-xiii-self-acceleration-exponent.

## 15 The growth equation THE DIFFERENTIAL EQUATION GENERATING EQ-05

**In words.** The rate at which capability improves per unit of recursive depth is proportional to current capability raised to the coupling power.

$$dg/dr = a * g^{(\beta_L)}$$

**What it is for.** The mechanism. Solve it and EQ-05 falls out. It is the only place the self-reference is actually stated as a process rather than a result.

**Measured as.** Varied: recursive depth r. Read out: the rate of change of capability g with depth. Held fixed: the configuration ladder and the scoring instrument, because beta\_L is claimed constant only over the range studied and a changed instrument would hide that.

$g$  growth variables of the differential equation

$r$  growth variables of the differential equation

$a$  growth variables of the differential equation

In the growth equation,  $g$  is current capability,  $r$  is depth, and  $a$  is a constant of proportionality that sets the pace but not the shape.

**How it is measured.**  $g$  and  $r$  as for  $U$  and  $R$ . The constant  $a$  is fitted and carries no claim: it moves the curve up or down without changing the exponent, which is what the programme is about.

**Domain.** positive

**Units.** as for  $U$  and  $R$ ;  $a$  carries the units that balance the equation

$\beta_L$  ( $\beta_L$ ) the self-referential coupling

Defined in full at EQ-05, its first use.

**Status.** standard separable ODE; the content is in the claim that  $\beta_L$  is constant over the range studied

Relied upon in: paper-iii-alignment-scaling-problem, paper-ix-synthesis-and-roadmap, paper-xiii-self-acceleration-exponent.

## 16 Error decay

**In words.** Error falls away as recursive depth rises, as a power of that depth.

$$E(R) = E_0 * R^{(-\alpha)}$$

**What it is for.** The measured face of EQ-01. What EQ-07 is fitted to.

**Read this before using it.** The corpus writes the exponent here as both  $\alpha$  and  $\alpha_{\text{align}}$ . If they are the same quantity, one symbol should be used; if they are different, the difference has never been stated. Open notation defect, logged in collisions.

**Measured as.** Varied: recursive depth  $R$ . Read out: error  $E$ . Held fixed: the error instrument, and the starting value  $E_0$ , which is the first measured point of the series being fitted rather than a free parameter.

$E$  error

Defined in full at EQ-07, its first use.

$E_0$  ( $E_0$ ) the starting values

Where alignment and error stood before the growth being described began. Every power law needs a starting point, and this is it.

**How it is measured.** The value of alignment, and of error, at the first measured point of the series being fitted.

**Domain.** positive

**Units.** as for the quantity each scales

$R$  recursive depth

Defined in full at EQ-01, its first use.

$\alpha$  (alpha) the capability-growth exponent

Defined in full at EQ-01, its first use.

**Status.** measured form

Relied upon in: paper-ii-experimental-validation, paper-iii-alignment-scaling-problem.

## 17 The saturation curve

**In words.** Alignment rises with capability but flattens towards a maximum, with the steepness set by an exponent.

$$A(C) = A_{\max} * C^{\beta_H} / (C^{\beta_H} + C_s^{\beta_H})$$

**What it is for.** Models alignment that cannot exceed a ceiling, unlike a bare power law which rises without limit.

**Read this before using it.** The exponent here is a steepness parameter of a saturating curve, a Hill coefficient. It is the third distinct quantity in the corpus written with the letter beta. See collisions.

**Measured as.** Varied: capability C. Read out: alignment A. Held fixed: the alignment instrument and the capability scale.  $A_{\max}$  and  $C_s$  are fitted constants of the curve, not measured quantities.

**A** alignment

How well the system's behaviour matches what was intended.

**How it is measured.** Blinded score under an audited consensus protocol, with model-level self-exclusion, six to seven blinded scores per entry.

**Domain.** bounded above by  $A_{\max}$  where a saturating form is used

**Units.** score, dimensionless once normalised

**C** capability

How capable the system is, used where capability rather than recursive depth is the driving variable.

**How it is measured.** Battery score, as for U.

**Domain.** positive

**Units.** same as U

**$A_{\max}$**  the saturation constants

**$C_s$**  ( $C_s$ ) the saturation constants

The most alignment a system can reach however capable it becomes, and the capability at which it is halfway there.

**How it is measured.** Fitted constants of the saturation curve: the upper asymptote, and the capability at half that asymptote.

**Domain.** both positive

**Units.** A\_max dimensionless; Cs in capability units

$\beta_H$  (beta\_H) the saturation steepness

How sharply alignment flattens out as it approaches its maximum. A high value means a sudden knee in the curve; a low value means a gentle bend.

**How it is measured.** Fitted parameter of the saturation curve EQ-09. Not an exponent of a power law and not comparable with the other betas.

**Domain.** positive

**Status.** modelling choice, not derived

Relied upon in: paper-x-coupled-coscaling.

---

## 18 Alignment scaling

**In words.** Alignment grows as a power of recursive depth.

$$A(R) = A_0 * R^{\beta_A}$$

**What it is for.** The alignment analogue of EQ-01.

**Read this before using it.** Fourth distinct beta. See collisions.

**Measured as.** Varied: recursive depth R. Read out: alignment A. Held fixed: the alignment instrument, and the starting value A<sub>0</sub>.

A alignment

Defined in full at EQ-09, its first use.

A<sub>0</sub> (A0) the starting values

Where alignment and error stood before the growth being described began. Every power law needs a starting point, and this is it.

**How it is measured.** The value of alignment, and of error, at the first measured point of the series being fitted.

**Domain.** positive

**Units.** as for the quantity each scales

R recursive depth

Defined in full at EQ-01, its first use.

$\beta_A$  (beta\_A) the alignment-scaling exponent

How alignment grows with recursive depth, as a growth-shape number of its own.

**How it is measured.** Log-log slope of alignment score against recursive depth.

**Domain.** real

**Status.** measured form

Relied upon in: paper-ii-experimental-validation.

---

## 19 The self-acceleration exponent

**In words.** How fast a system's improvement accelerates equals its cost-scaling exponent minus one over its growth-shape number.

$$k = \text{delta} - 1/\alpha$$

**What it is for.** Supplies the drift side of EQ-04 from measurable quantities, so the safety criterion does not need a separate instrument.

**Measured as.** Not measured directly. It is computed from two measured exponents, the cost-scaling exponent delta and alpha from EQ-07, so its interval inherits both of theirs and must be reported with them.

k the drift-acceleration exponent

Defined in full at EQ-04, its first use.

$\delta$  (delta) the cost-scaling exponent

How the cost of each further improvement grows as the system gets better. At zero, every step costs the same.

**How it is measured.** Log-log slope of per-step cost against capability.

**Domain.** real

$\alpha$  (alpha) the capability-growth exponent

Defined in full at EQ-01, its first use.

**Status.** derived in Paper XIII; an earlier form  $k = \beta_L - 1$  was withdrawn in full

Relied upon in: paper-xiii-self-acceleration-exponent.

---

## 20 The boundary gap

**In words.** The gap between two systems' shortfalls, expressed as a ratio.

$$\text{beta\_X} = (1 - \text{beta\_L}) / (1 - \text{beta\_L\_prime})$$

**What it is for.** Compares two self-referential systems on one scale.

**Read this before using it.** Fifth distinct beta, and the only one that is a ratio rather than an exponent. Paper XIII subscripts it correctly; no other paper adopted the convention.

**Measured as.** Not measured directly. It is a ratio of two shortfalls, each computed from a beta\_L measured on its own system through EQ-06, so the two systems must be measured on the same instrument before the ratio means anything.

$\beta\_X$  (beta\_X) the boundary gap

A way of comparing two self-improving systems on one scale, by dividing one system's shortfall by the other's.

**How it is measured.** Computed from two fitted couplings through EQ-12. A ratio, not an exponent.

**Domain.** at most one

**Units.** dimensionless ratio

$\beta\_L$  (beta\_L) the self-referential coupling

Defined in full at EQ-05, its first use.

$\beta\_L'$  (beta\_L\_prime) the comparison coupling

The same self-reference number measured on a second system, so two systems can be compared rather than one described.

**How it is measured.** The self-referential coupling of the comparison system in the boundary-gap ratio.

**Domain.** less than one

**Status.** derived

Relied upon in: paper-xiii-self-acceleration-exponent.

---

## 21 The metabolic analogy

**In words.** Power per unit mass falls as mass rises, to the power minus one quarter.

$$P/M = M^{(3/4)} / M = M^{(-1/4)}$$

**What it is for.** The biological precedent for a scaling exponent that is argued for from structure rather than fitted to data. Kleiber's three-quarter exponent is the observation; the network derivation offered for it by West, Brown and Enquist is the argument this analogy points at, and it is disputed: a competing account derives two thirds from surface-to-volume scaling, and the empirical exponent is not settled between them. What that literature holds and this programme does not is a mechanism grounded in measured anatomy. The analogy is used here only to show that a derived exponent is a thing science already has, and nothing in this programme depends on which exponent wins. Illustrative, not load-bearing.

**Measured as.** Not measured by this programme. Mass and metabolic power are measured in comparative physiology, and the result is carried here as an illustration; nothing in this programme is fitted to it.

**P** mass and metabolic power, in the biological analogy

**M** mass and metabolic power, in the biological analogy

How heavy an organism is, and how much energy it burns. The analogy notes that bigger animals burn less energy per kilogram, which is a real scaling law from biology and is used here only as an illustration.

**How it is measured.** Standard biological notation. This P is metabolic power and is NOT the P of probability in EQ-19.

**Domain.** positive

**Units.** M in mass; P in power

**Status.** established result from the literature, cited not claimed

Relied upon in: on-the-origin-of-scaling-laws.

## Symbols this register defines that no equation here carries

8 symbols are defined in the notation spine and appear in none of the equations above. They are set out here rather than attached to an equation, because giving them a host equation would change what the register claims rather than only how it renders. Each carries its plain definition, its domain, and where in the corpus it is used.

$\gamma_N$  ( $\gamma_N$ ) the residual-decay exponent

How fast the mistakes that are left over disappear as you keep revising. Go round the loop again and again, and this number says how quickly what remains shrinks. Higher means the leftovers vanish faster.

**How it is measured.** The slope of residual error against revision count at fixed capability: residual error after N revisions falls as N to the power minus this exponent. Fitted by the depth studies. It appears in no equation in this register, and it has never been measured on any real system.

**Domain.** positive; no upper bound is claimed

**What this rests on.** The programme's working premise treats this exponent as the same quantity as the correction-leverage exponent, under an assumption that internally accumulated corrections combine like independent samples. That identification is a registered question, not a settled fact, and no result depends on it being true.

$\gamma_D$  (gamma\_D) the capability-to-depth conversion exponent

How readily raw ability turns into more rounds of self-improvement. If a system gets cleverer, does it also get faster at improving itself, and by how much.

**How it is measured.** The exponent by which revision depth accelerates with capability: the rate of change of depth grows as capability to the power of this exponent. Not measured by any registered study.

**Domain.** positive

**What this rests on.** This exponent carries its own finite-time ceiling, one divided by it, and that is a DIFFERENT law from the ceiling law EQ-03. The two agree at exactly one half and nowhere else, which is why the form  $1/\gamma$  can look correct beside a value of two. Any sentence pairing this exponent's number with EQ-03's formula is false everywhere except that single point.

$\Delta$  (Delta\_balance) the balance exponent

Whether correction is pulling ahead of the work it has to do, read along the path a system actually took. Positive means the gap is closing.

**How it is measured.** The difference of two log-derivatives taken along an observed path:  $\phi_Q$  minus  $\phi_W$ . Directly estimable from the path alone; its components are not. Introduced in the author-review working paper of 16 August 2026.

**Domain.** any real number

**Where it is used.** the balance model; positive means correction is gaining on burden

$\phi_Q$  (phi\_Q) the correction-service path elasticity

How fast the correction a system can actually deliver grows as it goes deeper, measured along the path it took.

**How it is measured.**  $d \log Q / d \log r$  along the observed path, where  $Q$  is delivered correction service. Equals  $\alpha_C$  times  $\chi$  plus  $\eta$  when the path is decomposed, but is estimable without that decomposition.

**Domain.** any real number

**Where it is used.** the balance exponent

$\phi_W$  (phi\_W) the burden path elasticity

How fast the correctable work a system generates grows as it goes deeper, measured along the path it took.

**How it is measured.**  $d \log W / d \log r$  along the observed path, where  $W$  is newly generated correctable burden. Equals  $\alpha_C$  times  $\kappa$  plus  $\zeta$  when decomposed, but is estimable without that decomposition.

**Domain.** any real number

**Where it is used.** the balance exponent

$\kappa$  (kappa) the burden intensity elasticity

How much more correctable work each unit of new capability brings with it, as the system gets more capable.

**How it is measured.** The elasticity of newly generated burden with respect to capability. The registration's burden model is the case  $\kappa$  equals one; the burden-intensity rival is  $\kappa$  above one, written elsewhere as one plus  $\mu_B$ .

**Domain.** non-negative

**Where it is used.** the general ceiling; the burden-intensity rival

$\eta$  (eta) the direct depth-dependence of correction service

Whether correction gets better simply by being run more times, apart from any gain in capability.

**How it is measured.** The direct exponent of recursive depth in correction service, holding capability fixed. Zero in the registration's model. Embedded correction is the claim that this is larger than it is for correction applied from outside.

**Domain.** any real number

**Where it is used.** the general ceiling; the embedded-correction claim

$\zeta$  (zeta) the direct depth-dependence of burden

Whether going deeper generates correctable work by itself, apart from the capability it produces.

**How it is measured.** The direct exponent of recursive depth in newly generated burden, holding capability fixed. Minus one in the registration's model, which is what it means for burden to follow the rate of capability gain rather than its level.

**Domain.** any real number

**Where it is used.** the general ceiling

## Where one letter means two things

A symbol that carries two meanings is the failure this page exists to prevent. Both readings are named; neither is quietly dropped.

---

### gamma: 3 distinct quantities

- $\gamma$  the correction-leverage exponent, also called the correction-capacity exponent; the one in the ceiling law. Written  $\chi$  from the working paper of 16 August 2026 onward, which is the name to use in any new writing; the registration still says  $\gamma$  and the two are the same quantity (EQ-03)
- $\gamma_N$  the residual-decay exponent, how fast leftover error falls with revision count at fixed capability (the depth studies; no equation in this register)
- $\gamma_D$  the capability-to-depth conversion exponent, whose own finite-time ceiling is one divided by it (no equation in this register)

**Why it matters.** This is the collision with the worst history in the programme. The form  $\alpha_{\text{crit}} = 1/\gamma$  was retracted on 16 August 2026 for the correction-leverage reading, where it runs backwards. The SAME form is correct algebra for the conversion reading, because a different dynamical system sits underneath it. The two ceilings agree at exactly one half, which is the programme's headline number, so a wrong pairing survives casual review. Anyone writing  $\gamma$  without saying which of the three they mean can produce a sentence that is true under one reading, retracted under another, and unfalsifiable as written. Say which one, every time. One further reading exists and it is a rename rather than a fourth quantity: the ceiling-law reading is written  $\chi$  from 16 August 2026, because  $\gamma$  could no longer be written safely. The  $\chi$  entry says so; this entry now says it too, so the pointer runs both ways and a reader meeting either symbol finds the other.

**Resolution.** Write the full name in prose on first use in any document, and use a subscript only alongside it. The bare letter  $\gamma$  is reserved for the correction-leverage exponent of EQ-03 and means nothing else.

---

### beta: 5 distinct quantities

- $\beta_C$  correction exponent, Law II, compared against  $k$  (EQ-04)
- $\beta_L$  self-referential coupling, Foundational Theorem 2, giving  $\alpha = 1/(1-\beta_L)$  (EQ-05, EQ-06)
- $\beta_H$  saturation steepness, a Hill coefficient (EQ-09)
- $\beta_A$  alignment-scaling exponent (EQ-10)
- $\beta_X$  boundary gap, a ratio of shortfalls (EQ-12)

**Why it matters.** A reader or a machine that conflates  $\beta_C$  with  $\beta_L$  gets the safety criterion exactly backwards, because one is a correction rate compared against drift and the other is a coupling that feeds a reciprocal. Paper X uses a bare  $\beta$  for two different quantities within one paper.

**Resolution.** This file subscripts all five. The corpus has not yet adopted the subscripts outside Paper XIII, which introduced them independently and correctly. Propagation across the papers is outstanding, and it belongs to the papers rather than to this register: this file records the convention, and each paper adopts it at its next version.

---

### alpha in the error-decay equation: 2 distinct quantities

- $\alpha$  the capability-growth exponent (EQ-08 as written in some papers)
- $\alpha_{\text{align}}$  an alignment-specific decay exponent (EQ-08 as written in others)

**Why it matters.** If these are the same quantity, one symbol should be used. If they are different, the difference has never been stated anywhere in the corpus.

**Resolution.** Unresolved. Flagged for the papers to settle, not settled here, because settling it is a scientific judgement and this file records judgements rather than making them.

---

### **d: 3 distinct quantities**

- $d$  effective spatial dimension, in  $\alpha = d/(d+1)$  and supplied by the equation-of-state relation  $d = 2/(1+3w)$ ; one quantity across two equations, not two (EQ-13)
- $d$  (Cohen) Cohen's  $d$ , a standardised effect size, reported as  $d = -0.53$ ; a statistic about a comparison, carrying no relation to dimension (results reported across seven papers)
- $d(t)$  the misalignment fraction, defined as  $D/C$ , how much of a system's capability is misaligned; a bounded ratio that evolves in time (Paper X)

**Why it matters.** A reader meeting  $d$  in a results table and  $d$  in the dimensional derivation is meeting two unrelated quantities, and  $d(t)$  in Paper X is a third. Cohen's  $d$  is standard notation nobody should change, which makes the collision permanent rather than fixable: the answer is to subscript the programme's own uses, never the statistic.

**Resolution.** Unresolved in the corpus. This file records all three. The dimensional  $d$  and the misalignment fraction  $d(t)$  should carry distinguishing marks in the papers; Cohen's  $d$  must be left exactly as it is, because it belongs to the discipline rather than to this programme.

---

### **A: 2 distinct quantities**

- $A$  alignment, the degree to which behaviour matches intent (EQ-09, EQ-10)
- $A_c$  correction capacity, how much correcting a system can do, the denominator of the burden ratio (EQ-15)

**Why it matters.** The burden-to-capacity ratio is the derivation the ceiling comes from, so confusing its denominator with alignment misreads the origin of the value two.

**Resolution.** Subscripted as  $A_c$  in this file. Introduced here rather than inherited: the papers write a bare  $A$ . Recorded as a collision this register CREATED by naming a quantity that had none, rather than one it found, because reusing the bare letter while auditing letter reuse would have been the same mistake in a new place.

---

### **P: 2 distinct quantities**

- $P$  probability, in Bayes' theorem (EQ-19)
- $P$  metabolic power, in the biological scaling analogy (EQ-14)

**Why it matters.** The two are unrelated and sit in equations of different kinds, one statistical and one illustrative. The risk is low but a reader meeting P for the first time in EQ-14 will carry the wrong meaning into EQ-19.

**Resolution.** Read P as probability everywhere except the metabolic analogy, which says so in words.

---

### delta and Delta: 3 distinct quantities

- delta (cost) the cost-scaling exponent (this register)
- delta\_obs (direct-bound unit) observed correction pressure, alpha times one minus the correction elasticity, minus one; POSITIVE when burden outruns correction (the direct-bound draft's H2)
- Delta (balance) phi\_Q minus phi\_W; POSITIVE when correction gains on burden (the balance model of 16 August 2026)

**Why it matters.** The two pressure quantities carry OPPOSITE sign conventions and are equal in magnitude at the registered dials: Delta equals minus delta\_obs there. A reader who meets both and assumes one convention reads a stability verdict backwards. Always name which is meant.

**Resolution.** Name the quantity, never the letter. Bare delta is the cost-scaling exponent and nothing else. The direct-bound unit's pressure reading is written delta\_obs in full and is POSITIVE when burden outruns correction; the balance model's reading is written Delta in full and is POSITIVE when correction gains on burden. Because the two are equal in magnitude and opposite in sign at the registered dials, a sign quoted without its symbol's full name decides a stability verdict the wrong way round, so neither may be quoted bare, and any passage carrying both states the convention it is using in the same sentence.

---

### E: 2 distinct quantities

- E error, how wrong the system is (EQ-07, EQ-08, EQ-18)
- E (Bayes) evidence, the observation being conditioned on (EQ-19)

**Why it matters.** E is defined as error five sections before it appears as evidence, so a reader carrying this file's own definition into Bayes' theorem reads the probability of a hypothesis given an error.

**Resolution.** Read E as error everywhere except inside Bayes' theorem, which is standard statistical notation and must not be renamed.

---

## **k: 2 distinct quantities**

- k the drift-acceleration exponent, compared against beta\_C (EQ-04, EQ-11)
- k (Fisher) the number of pooled p-values, setting the degrees of freedom (EQ-21)

**Why it matters.** 2k degrees of freedom is a count, not an exponent, and the exponent reading is the one this file defines first.

**Resolution.** The bare k is the drift-acceleration exponent; Fisher's k is written in words as the number of studies pooled.

## **Forms that were withdrawn**

Kept on the record rather than deleted. A programme that removes its own retractions loses the thing that made the correction honest.

---

**alpha\_crit = 1 / gamma**

**Replaced by**  $\alpha_{\text{crit}} = 1 / (1 - \gamma)$ , 2026-08-16.

**Why it was wrong.** It runs backwards under the programme's own definition of gamma: it makes a better corrector lower the ceiling. Under the same-class cap gamma at most one half it returns a ceiling of AT LEAST two, which inverts a bound into a floor.

**Why it went unnoticed.** The two forms return the same answer, two, at exactly gamma = one half, which is the programme's headline number. Agreement at the one point anybody checks.

Legitimate where: Inside a correction note, a version history, or a unit whose subject is discriminating the two forms. ALSO legitimate, and this is the exception that caused the confusion, as the finite-time ceiling of the capability-to-depth CONVERSION exponent gamma\_D, which is a different quantity obeying a different law. When used that way it must be written with the conversion symbol and never with a bare gamma, because the retraction governs the correction-leverage reading absolutely..

---

**k = beta\_L - 1**

**Replaced by**  $k = \delta - 1/\alpha$ , withdrawn in full in an earlier draft of Paper XIII.

**Why it was wrong.** Recorded as withdrawn in the paper itself; the replacement derives k from cost scaling rather than from the coupling.

Legitimate where: inside Paper XIII's own withdrawal note.

---

**alpha = 2.24 as a measured value**

**Replaced by** alpha = 0.49, interval [-1.3, 2.9], which distinguishes nothing, after cross-architecture replication failure.

**Why it was wrong.** It did not replicate across architectures.

Legitimate where: only ever as retracted; the bare figure is banned on every surface.

---

**alpha\_max**

**Replaced by** alpha\_crit, 2026-08-30.

**Why it was wrong.** The ceiling is a crossover on stability, not a maximum: a system can exceed it and cannot stay correctable there. alpha\_max reads as an impossibility limit, which the stability framing forbids, and it named the same quantity as alpha\_crit under a second symbol, so the glossary and the founding-paper drafts carried the law under two names.

**Why it went unnoticed.** Both symbols sat beside the same formula,  $1/(1 - \gamma)$ , so a reader never met the two at once and every surface hand-wrote whichever it had seen last.

Legitimate where: inside a correction note, a version history, or a passage discriminating the two symbols.

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**Law III, the ARC Bound**

**Replaced by** Law III, the ARC Ceiling, 2026-08-30.

**Why it was wrong.** The ARC Bound ( $\alpha \leq 2$ ) is the value the ARC Ceiling returns at  $\gamma = 1/2$ , not the law; naming the instance as the law hides the assumption the value rests on.

Legitimate where: inside a dated correction row or version note; as printed in the DOI-frozen papers, which carry the canonical name in their placement notes.

---

**Law III, the Ceiling**

**Replaced by** Law III, the ARC Ceiling, 2026-08-30.

**Why it was wrong.** The programme prefix is part of the law's name; the bare word is a description, and a description can attach to any ceiling.

Legitimate where: inside a dated correction row or version note; as printed in the DOI-frozen papers.

---

## the Co-Scaling Law

**Replaced by** the ARC Co-Scaling Law, 2026-08-30.

**Why it was wrong.** Law II's name carries the programme prefix like the other two; the bare form drifted onto surfaces written from memory.

Legitimate where: inside a dated correction row; as printed in the DOI-frozen papers, whose placement notes carry the canonical name.

## What is still open

- Is the exponent in the error-decay equation the same  $\alpha$  as the capability-growth exponent, or a distinct alignment-specific one? Never stated.
- The independence premise behind  $\gamma = \text{one half}$  is undefended. It is the single assumption holding up the value two, and it is deliberately left open.
- EQ-05 supplies an unexploited falsifier: measure  $\beta_L$  independently, predict  $\alpha$ , compare against the fitted  $\alpha$ . No unit currently does this.
- Cohen's  $d$  and the programme's dimensional  $d$  share a letter and cannot both be renamed, because one belongs to statistics rather than to this programme. Which of the programme's own uses takes a subscript is an open decision.
- Does build order actually change the corrector's class enough to move the leverage exponent? The argument says order is a lever; whether pulling it separates the classes in practice is unmeasured, and it is the cheapest decisive study the programme has specified.

## Version history

Version 1.9.0 (4 October 2026): published as a paper in every format, with its own DOI on OSF and its own reference card, under the title Operational Definitions of the ARC Theory (until this version, Operational Definitions of the ARC Programme). The note on what the word law is doing counts the theory's five laws and their status, and the description and the abstract say where the notation of Laws IV and V is fixed. No equation, symbol, collision or retracted form changes.

Version 1.8.0 (7 September 2026): thirty-three of the fifty-one individual symbol names printed in these equations had no rendered definition block, because the renderer looked names up in a dictionary keyed by the spine's full composite strings and silently dropped anything it did not find, because six equations did not name in symbols\_ used the symbols their own statement prints, and because eight symbols appear in no equation at all. Every symbol's domain and units were carried in this file and rendered nowhere. The lookup now fails loudly, the composite keys resolve member by member, the orphans have a standalone section, the domain and units render, and the gate tests for a rendered definition block rather than for a name occurring somewhere on the page, which 46 of the 51 already did. It supersedes 1.7.2 (sha256 0ae774690effd36e9c19eafee863300dfce38ca2f2687ffa20a969923af927), not 1.7.1. 1.7.2 (2026-09-04): the delta family was declared a collision with readings, a severity and a why-it-matters and no resolution key, so build-operationalisation.py raised `KeyError('resolution')` and the production chain died on it. The resolution now names the quantity rather than the letter: bare delta is the cost-scaling exponent and nothing else, the direct-bound unit's pressure reading is written `delta_obs` in full and is positive when burden outruns correction, the balance model's reading is written `Delta` in full and is positive when correction gains on burden, and because the two are equal in magnitude and

opposite in sign at the registered dials neither may be quoted bare. One field was added to one collision in commit 68bbb55f91; no equation, symbol, definition or count changed, and the register still carried 21 equations, 38 symbols, 7 collisions, 7 retracted forms and 5 open questions after it as before. Those bytes hash to sha256 oae774690effd36e9c19eafee863300dfce38ca2f2687ffaae20a969923af927 and stood under the number 1.7.1 from 4 September 2026 until the 1.8.0 bump. They are given their own number here, in the record and not by editing them, so that 1.7.1 names one set of bytes and one only: 1.7.1 (2026-09-03) is the deposited state, sha256 68891fa12cf3ee4a58d1a7d278aecdd9ed170366339e36cc7ab478027bbbeb48, which the theory registration's companion extract of 3 September 2026 cites and which nothing in this record changes. 1.7.0 (2026-09-03): the gamma collision, which is the register's worst, pointed one way only. The chi entry recorded that it is the registration's gamma; the gamma entry did not record that the quantity is now written chi, so a reader arriving at the collision was not told the current name existed. 1.6.0 (2026-08-27): worked example ran a bounded percentage through unbounded arithmetic; recast on the ladder-scale latent measure with the three model families stated (27 August 2026); then, 30 August 2026, the law labels were aligned with the home declaration (Law II is the ARC Co-Scaling Law; Law III is the ARC Ceiling, of which the ARC Bound is the same-class instance) and alpha\_max was retired as a second symbol for alpha\_crit

First published on 22 August 2026; versions 1.5.0 (24 August), 1.6.0 to 1.6.2 (27 to 30 August) and 1.7.0 to 1.7.1 (3 September) built and corrected the spine, each dated by its commit.

## How to cite this paper

DOI: 10.17605/OSF.IO/GVQ82

APA (AUTHOR-DATE, 7TH EDITION)

Eastwood, M. D. (2026). Operational Definitions of the ARC Theory.  
<https://doi.org/10.17605/OSF.IO/GVQ82>.  
<https://www.michaeldariuseastwood.com/research/papers/operational-definitions.html>

BIBTEX

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  doi = {10.17605/OSF.IO/GVQ82},  
  url =  
{https://www.michaeldariuseastwood.com/research/papers/operational-definitions.html}  
}
```

INLINE

Eastwood, 2026

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*Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.*